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Venture Viability Analysis

srisairam engineering college

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We develop a smart mobility app that provides real-time bus tracking and live crowd density updates



srisairam engineering college

Venture Team



Name: J SHOPHANA
Major: Engineering



Name: S.G Pushpasree
Major: Engineering



Name: B.Monisha
Major: Engineering



Context

tudies show that passengers often perceive bus waiting times to be longer than the actual waiting ti

Problem Statement

Problem

Bus passengers, students, and working professionals face uncertainty about bus arrival times and locations, leading to long waiting periods, missed bu

Impact

Passengers face long waiting times and missed buses, students arrive late to classes, working professionals experience delays and reduced productivity



Problem Statement/Industry

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Problem Being Solved

"Passengers, students, and working professionals often face uncertainty about the real-time location and arrival time of buses. This lack of accurate and timely information leads to long waiting times, missed buses, delays in reaching destinations, and difficulty in planning daily travel. Transport operators also struggle to efficiently monitor bus

Supporting Data

tudies show that passengers often perceive bus waiting times to be longer than the actual waiting ti

Source: The data is sourced from the research paper “Where Is My Bus? Impact of Mobile Real-Time Information



Area
Mobility



Industry
Transportation & Logistics



Domain
Public Transporation

Problem Analysis

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Affected Stakeholders

us passengers and daily commuters – face uncertainty about bus arrival times and delays. Students – may reach schools and colleges late due to unpredictable bus schedules. Working professionals – experience delays in commuting to workplaces. Bus operators and transport companies – find it difficult to monitor bus movements and service performance.



Impact on Stakeholders

Bus passengers and daily commuters – Experience long waiting times, missed buses, and difficulty planning their travel. Students – May arrive late to classes, affecting attendance and academic performance. Working professionals – Face delays in reaching workplaces, reducing productivity and causing stress. Bus operators and transport companies – St



Root Causes

Lack of real-time tracking and monitoring of buses. Traffic congestion causing unpredictable delays. Inaccurate or outdated bus schedule information. Poor communication between transport operators and passengers. Limited visibility of bus locations for commuters. Inefficient route and fleet management by transport authorities.



Personal/Team Connect

As students, we regularly depend on public transportation for commuting to college and other places. We have personally experienced long waiting times, uncertainty about bus arrivals, and difficulties caused by delays. These challenges affect our daily schedules and productivity. Since our team has an interest in technology and problem-solving, we

Target Customer Segments

Primary

Urban bus passengers, students, and daily commuters who rely on public transportation for their ever



Secondary

college students & daily campus travel

Customer Segment & Persona

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Primary Segment

Urban bus passengers, students, and daily commuters who rely on public transportation for their ever

Secondary Segment

college students & daily campus travel

Persona



Arun kumar
Age in years: 21
Location: Rural
Organizational Role: {Persona's primary role}
(if applicable)

Customer Profile

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Education: Post-graduate

Gender: Male

Occupation: Student

Interests/Hobbies: Travel

Primary Source of Information: Social Media

Shopping Preference: Mostly Online

Comfort with Technology: High

Favourite Social Media: intagram

Favourite Offline Gathering Spots: shopping malls

Jobs-to-be-Done

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Functional JTBD



Functional Jobs-to-be-done * List practical tasks the persona wants to accomplish which the problem prevents them from. Example: As a student - understand concepts, get good grades

Emotional JTBD



Feel confident about reaching the destination on time. Feel stress-free while waiting for the bus. Feel in control of their daily travel schedule. Feel safe and comfortable during their commute. Avoid anxiety and frustration

Social JTBD



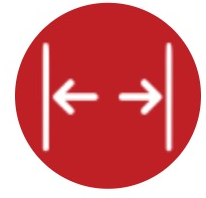
Social Jobs-to-be-Done Be seen as punctual and responsible by teachers and classmates. Avoid being late to college, classes, or events. Maintain a good reputation among friends and peers. Be able to plan outings and meetups with friends confidently. Receive appreciation from teachers and family for being punctual.

Current Alternatives



Current Alternatives

Current Solutions/Alternatives Checking fixed bus timetables Asking bus conductors or drivers Waiting at the bus stop and estimating arrival time Asking friends or other passengers Using Google Maps or existing transport apps Taking auto-rickshaws or taxis when the bus is delayed Calling family or friends to coordinate travel plans



Gaps in Current Alternatives

Limitations of Existing Solutions Bus timetables may be outdated or inaccurate. Passengers often cannot see the bus's exact location. Traffic and unexpected delays make arrival times unpredictable. Asking drivers or other passengers is not always reliable. Existing apps may provide limited or inaccurate information. Passengers may have to wait for

Problem Validation (GOOTB)

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Partial List of Potential Customers/Users Interviewed

Name: Arun Kumar

Occupation: College Student

Name: Priya

Occupation: College Student

Name: Karthik

Occupation: College Student

Problem Validation

Total customers/users interviewed:

- In-person: 10
- Virtually: 0

Total customers/users for whom this problem is important to solve: 8

Total customers/users who are dissatisfied with the current alternatives: 7

Our Solution



Solution

Real-Time Bus Tracking and Smart Crowd Management System - Develop an IoT and GPS-enabled mobile application that tracks live bus locations, estimates arrival times, and uses passenger counting sensor

Core Technologies/ Methodologies

GPS tracking technology, IoT passenger-counting sensors (IR/camera sensors), Flutter/React Native for mobile app development, Node.js/Python backend,

Solution Design

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Our Solution

Real-Time Bus Tracking and Smart Crowd Management System - Develop an IoT and GPS-enabled mobile application that tracks live bus locations, estimates arrival times, and uses passenger counting sensor

Key Features

Live GPS bus location tracking, real-time crowd density indicators (low, medium, high), estimated time of arrival (ETA) predictions, route search, and crowd alerts for commuters

Uniqueness

Integrated crowd density tracking paired with real-time GPS location, allowing passengers to plan trips based on passenger volume rather than just arrival time.

Solution Format:
Digital Product (Mobile Application and Cloud-base

Core Technologies/ Methodologies:
GPS tracking technology, IoT passenger-counting sensors (IR/camera sensors), Flutter/React Native for mobile app development, Node.js/Python backend,

Solution Benefits

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Functional Benefits

Accurate real-time bus location tracking, accurate arrival time predictions, and visibility into bus crowd density to avoid overcrowded buses and reduce waiting time



Emotional Benefits

Peace of mind regarding travel timing, reduced commute stress and anxiety, and a feeling of control over daily travel planning



Social Benefits

Punctuality for work and classes, safer and more comfortable travel experiences, and recognition among peers as a smart, tech-savvy commuter.



Macro Benefits

It improves public transport efficiency, reduces urban traffic congestion and vehicle emissions by encouraging public transit use, and saves commuters valuable daily time

Competitors



Direct

Chalo app where is my train,chartr app
Moovit, citymapper, transit app

Indirect

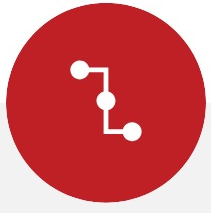
Google maps, uber,Ola,rapido
Grab,Gojek,Lyft

Our UVP

What makes your solution unique: Unlike basic GPS tracking apps that only show a bus's location, you

Competitors

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Direct Competitors

Chalo app where is my train,chartr app



Indirect Competitors

Google maps, uber,Ola,rapido



Direct Competitors Globally

Moovit, citymapper, transit app



Indirect Competitors Globally

Grab,Gojek,Lyft

Macro Analysis

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Favourable Trends

AREA	DESCRIPTION
Technology	Growth of AI, IOT, GPS and smart sensors for real time public transportation
Social	Increasing demand for safe reliable and less crowded public transportation

Unfavourable Trends

AREA	DESCRIPTION
Economy	Cost of GPS sensors and cameras communications cloud services and maintenance can increase implement
Technology	Poor connectivity can affect real time tracking while camera and sensor data can create privacy and

Data Sources:

Ministry of housing and urban affairs- smart cities mission NITI aayog report on artificial intelligence and digital technology Government of India guidelines on data privacy ,digital infrastructure.

Back-of-the-Envelope Financial Projections

Currency: Indian Rupee (INR)

Chosen Business Model: Subscriptions

AREA	YEAR 1	YEAR 2	YEAR 3
Revenues	600000	1200000	2100000
Total Expenses	800000	1200000	1560000
Profit	-200000	0	540000

Prototype

Prototype Format

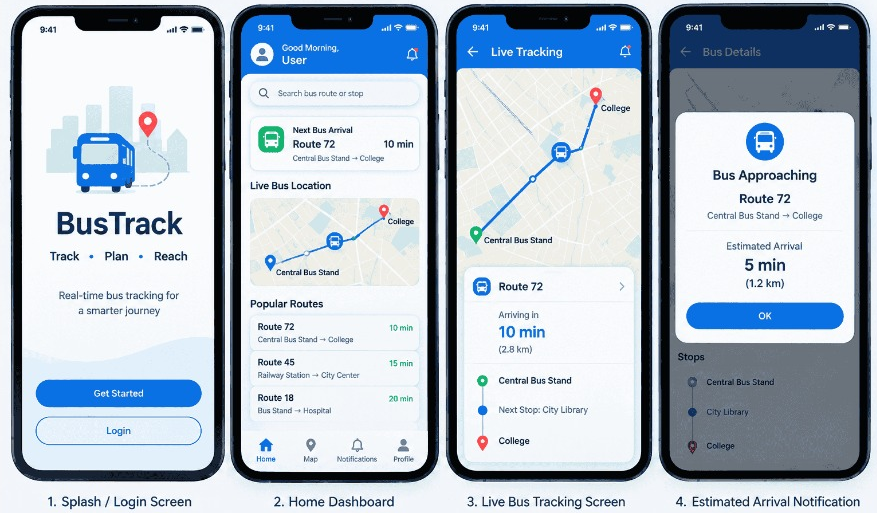
A digital mobile app prototype that demonstrates an AI-based bus tracking application. It will show

Functionality included in the Prototype

User-friendly home screen. Search for buses by route or bus number. Display bus location on a map using sample data. Estimated bus arrival Bus route and nearby bus stops. Delay notifications as a demo

Functionality NOT included in the Prototype

Live GPS tracking of actual buses. Integration with government transport databases. Fully functional AI-based arrival predictions. Real-time push notifications. Online payments and subscri



1. Splash / Login Screen
2. Home Dashboard
3. Live Bus Tracking Screen
4. Estimated Arrival Notification

{Upload another picture}

Prototype Validation

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Number of users engaged with?

10

How many people liked or loved the prototype?

8

How many people were either neutral or mostly unhappy with the prototype?

2

Prototype Feedback

What aspects of the prototype did the users LOVE?

Users liked the simple design, easy bus route search, map-based bus tracking, estimated arrival times, and clear display of nearby bus stops.

What aspects of the prototype were DISLIKED by the users?

Some users wanted more accurate arrival times, live GPS tracking, reliable delay notifications, and support for rural bus routes. They were concerned about inaccurate information.

Competition Analysis

COMPETITOR NAME	TYPE	STRENGTHS	WEAKNESSES
Chalo App	Direct	Provides live bus tracking	Rural route coverage may be limited
Google Maps	Direct	Well-known and trusted global brand	Bus arrival information depends on
Google Maps	Direct	Free to use and easy to access.	Some rural routes have limited info
Moovit	Direct	User-friendly mobile application.	Features and live tracking vary by

Our Product/Service will be better than the competitors' solutions because:

BusTrack will focus on rural and semi-urban bus routes, provide AI-based arrival time predictions using traffic and historical data, and offer a simple, affordable mobile app with low data usage and t

Market Size & GTM

GTM Channels

Digital

Instagram, YouTube, Facebook, WhatsApp Business, Google Ads, and LinkedIn.

Physical

Direct meetings with transport authorities, demonstrations at bus depots, college exhibitions, printed brochures, and transportation conferences.



Source: Astute Analytica 2025; Telematics Wire 2025

Market Size

Total Addressable Market (TAM)
50k crore annually, representing the estimated

Serviceable Available Market (SAM)
₹50 thousand

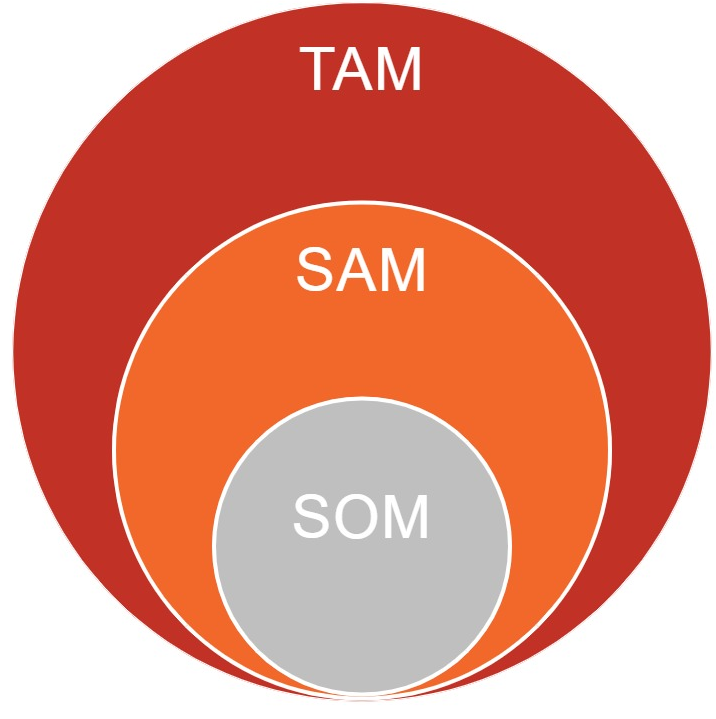
Serviceable Obtainable Market (SOM)
24lak

Assumptions

he service will initially target students, working professionals, and daily bus commuters in Tamil Nadu, especially in urban, semi-urban, and rural areas who need reliable bus arrival information.,
100,
1010

Sources of Research

Astute Analytica 2025; Telematics Wire 2025



Revenue Models / Pricing

Revenue Model (Primary)

Freemium Model

Revenue Model (Secondary)

Freemium Model

Lean Canvas

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Problem
Bus passengers, students, and working professionals face uncertainty about bus arrival times and locations, leading to long waiting periods, missed bu

Solution
Real-Time Bus Tracking and Smart Crowd Management System - Develop an IoT and GPS-enabled mobile application that tracks live bus locations, estimates arrival times, and uses passenger counting sensor

Key Metric
For Commuters / User Engagement: Daily Active Users (DAU), app retention rate, and search/track queries per day.

Unique value Proposition
What makes your solution unique: Unlike basic GPS tracking apps that only show a bus's location, you

Unfair Advantage
Proprietary Crowd-Density Algorithm: A unique data-analytics engine that accurately estimates and predicts crowd congestion levels per bus in real-time, which generic GPS apps cannot replicate easily.

Channels
Direct-to-Consumer (D2C) Mobile App Stores: Google Play Store and Apple App Store for the commuter-facing tracking app.

Customer Segment
Urban bus passengers, students, and daily commuters who rely on public transportation for their ever,college students & daily campus travel

Cost Structure
Technology & Infrastructure: Cloud hosting (servers/AWS), API integrations, map/GIS routing service fees, and ongoing software development and maintenance.

Revenue Streams
Freemium Model, Freemium Model

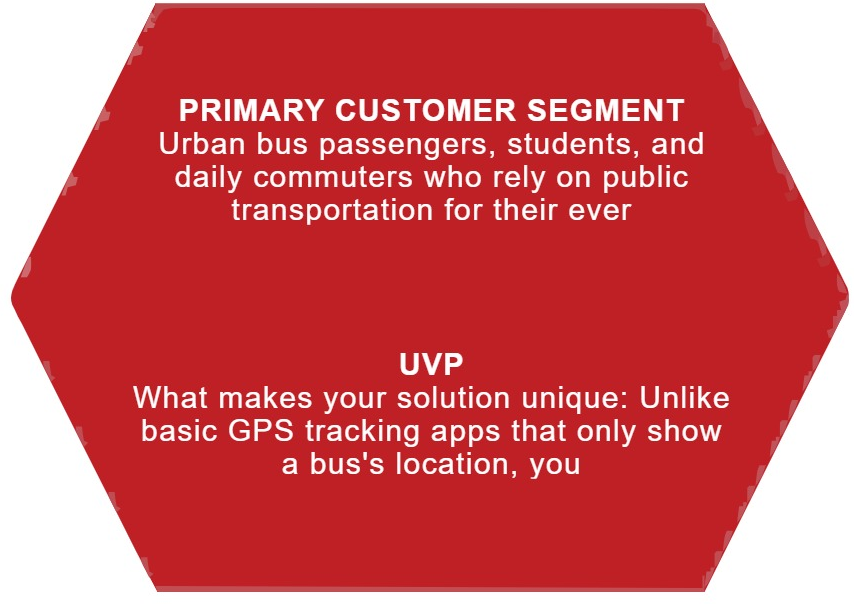
Go-to-Market Approach

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Geographic Focus
Chennai, Tambaram, Chengalpattu, Coimbatore, and Madurai.

Digital Marketing Channels
Instagram, YouTube, Facebook, WhatsApp Business, Google Ads, and LinkedIn.

Marketing KPIs
Number of buses tracked, passenger usage, customer satisfaction, app downloads, and revenue growth.



Physical Marketing Channels
Direct meetings with transport authorities, demonstrations at bus depots, college exhibitions, printed brochures, and transportation conferences.

Competitors' GTM
Competitors use websites, mobile apps, social media, and partnerships with transport authorities to promote their services.

GTM Partners
Government transport departments, private bus operators, colleges, technology companies, and GPS device providers.

Sales & Customer Service

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Customer Service
Customer support through phone calls, email, WhatsApp, and in-app help

Distribution Channels
Direct-to-Consumer (D2C) Mobile App Stores: Google Play Store and Apple App Store for the commuter-facing tracking app.

Physical Sales Channels
Direct meetings with transport authorities, bus depot demonstrations, college exhibitions, and printed brochures.

PRIMARY CUSTOMER SEGMENT

Urban bus passengers, students, and daily commuters who rely on public transportation for their ever

Digital Sales Channels
Company website, mobile app, Google Ads, social media, and WhatsApp Business.

Sales KPIs
Number of customers, buses tracked, app downloads, revenue growth, and new partnerships.

UVP:
What makes your solution unique: Unlike basic GPS tracking apps that only show a bus's location, you

https://drive.google.com/file/d/1Hev7E9eX2hjeFWGLKsqYOGEJvEamwp-c/view?usp=drivesdk

GTM Partners
Customer feedback forms, app ratings, online surveys, and complaints tracking.

https://drive.google.com/file/d/1Hev7E9eX2hjeFWGLKsqYOGEJvEamwp-c/view?usp=drivesdk

Financials

Revenue Models/Streams

- Freemium Model
- Freemium Model

Pricing

- **Unit of Sale:** Indian Rupee (INR)
- **Selling price per unit:** Subscription per bus per month

First Year Projections

Revenues: 10lakh

Operating Profits: 5laks

Revenue Models / Pricing

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Revenue Model (Primary)

Freemium Model

Unit of Sale

Indian Rupee (INR)

Sale Price per Unit

Subscription per bus per month

Expected units to be sold in Year 1

10

Expected growth in monthly sales

10% per month

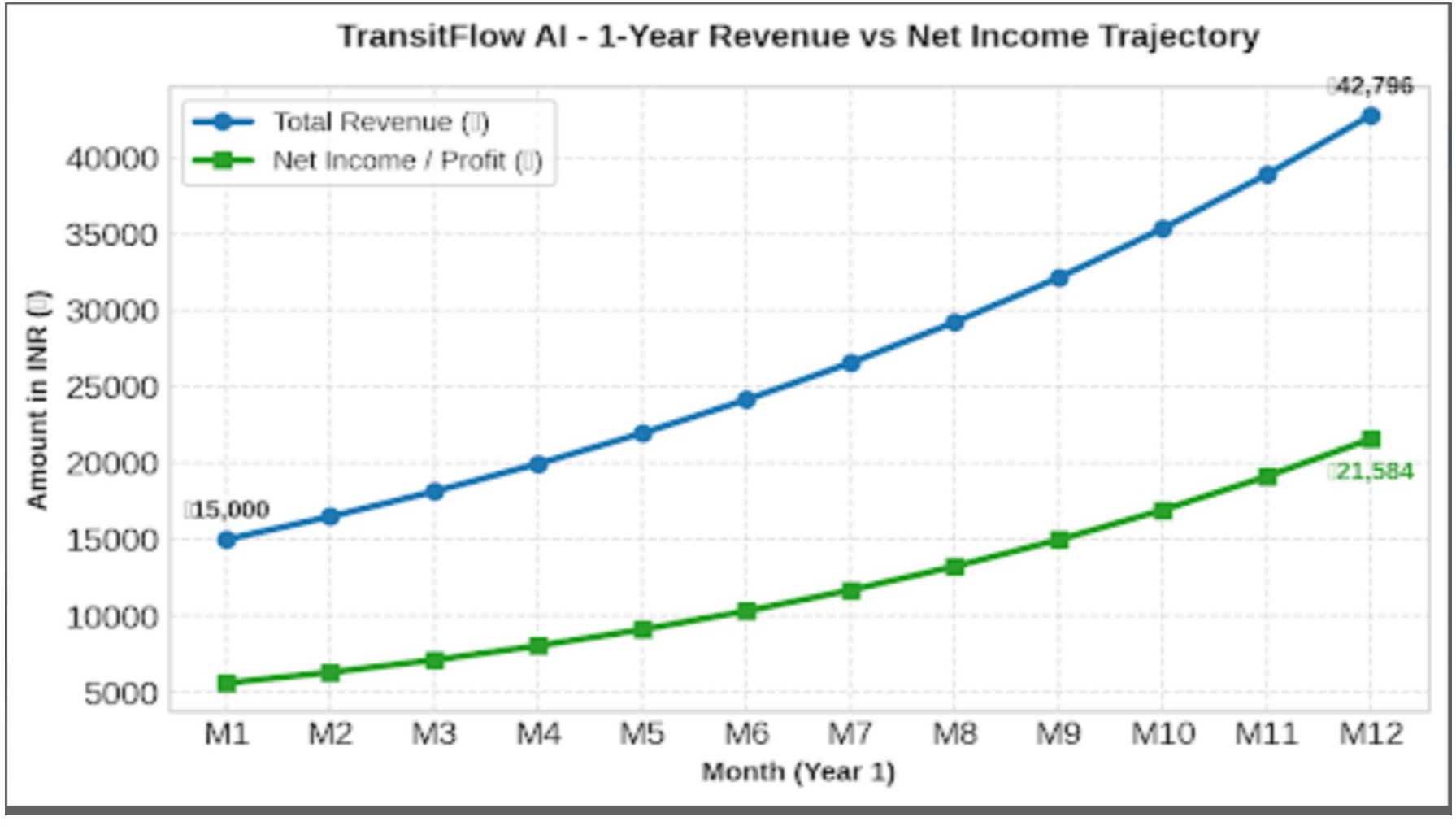
Costs & Revenues: Key Assumptions

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TransitFlow AI - 9.1 Assumptions for Financial Planning

Financial Assumption Parameter	Value / Assumption
Currency	Indian Rupee (INR / ₹)
Unit of Sale	Subscription per bus per month
Price Per Unit (Monthly)	₹1,500
First Month Projected Sales	10 Units (Buses)
Expected Monthly Sales Growth Rate	10% per month
Primary Target Customer	Private Bus Fleet Operators & Educational Institutions
Key Value Proposition	AI Real-Time Tracking & Dynamic Crowd Density Analytic

Profit & Loss Projections: Summary



Performance & Break-Even Analysis

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Year 1 Revenues

10lakh

Gross Profits for Year 1

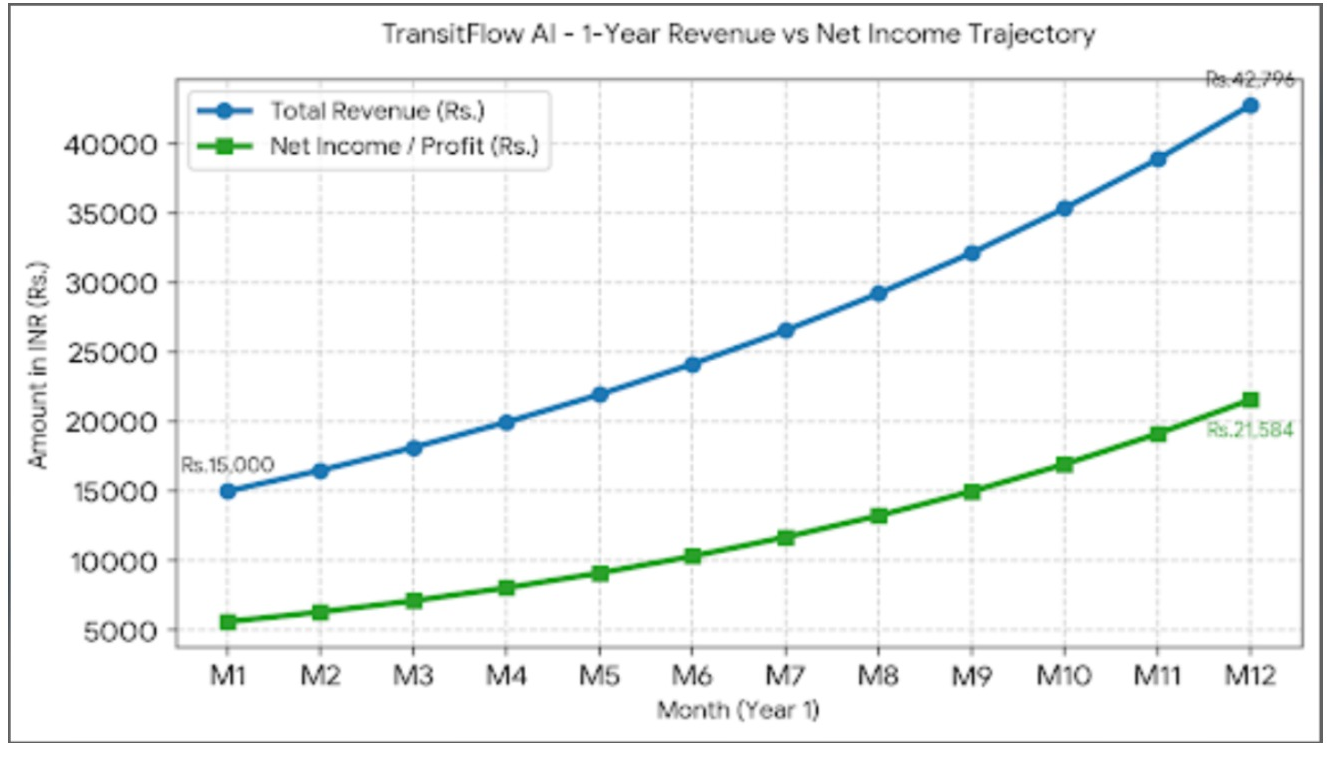
8 lakh

Net Profits for Year 1

5laks

Break-even Month

13 month



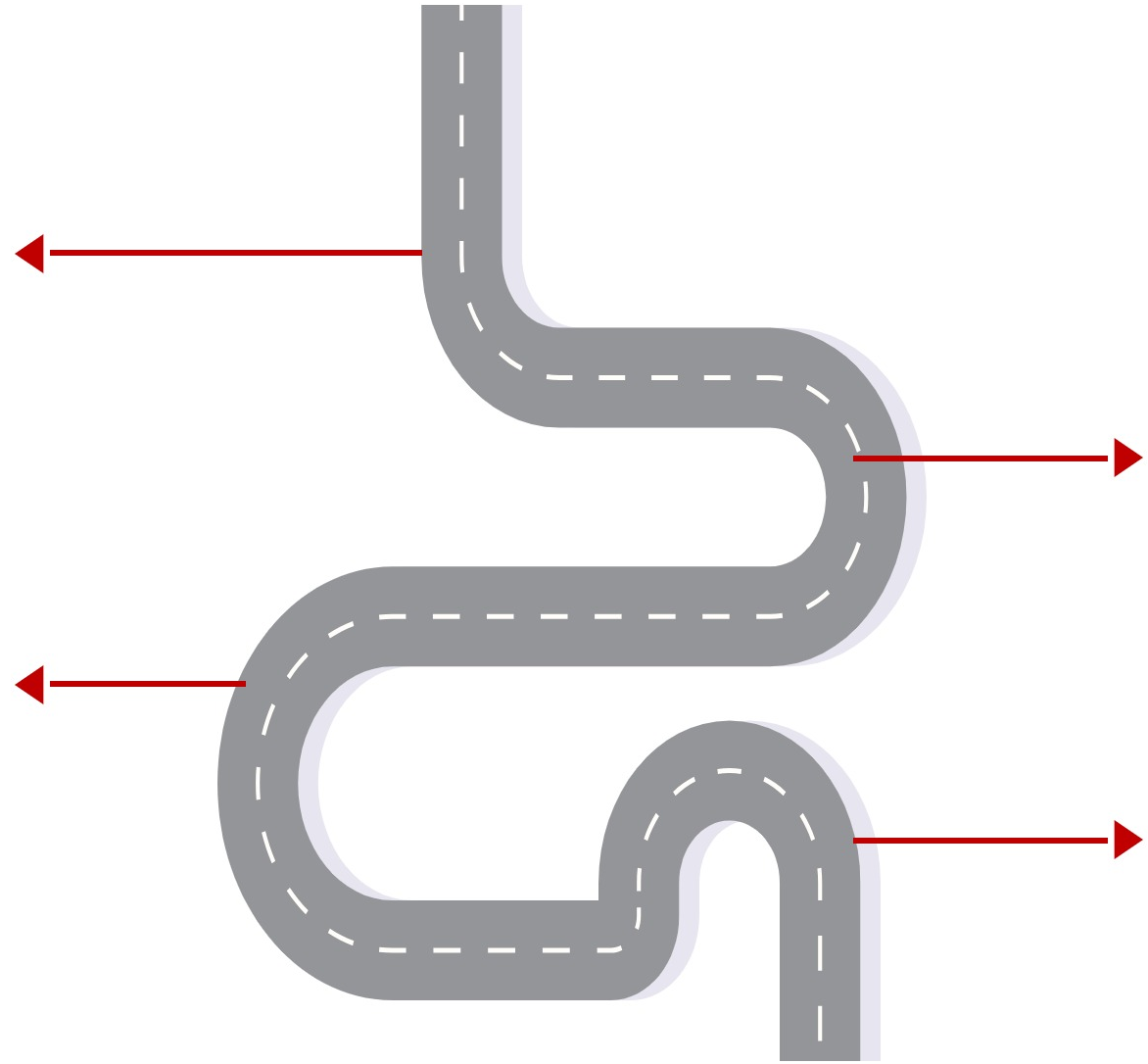
Next Steps

Goals for Months 10-12

Scale operations to 30+ buses

Goals for Months 4-6

Deploy pilot test on 10 buses and launch the pass



Venture Viability Index: 93.33%

Goals for Months 7-9

Validate pilot feedback, optimize crowd-counting

Goals for Months 1-3

Build functional working prototype and test real

Venture Viability Assessment

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Venture Viability Index

93.33%

Strengths

Our key strengths are a strong product value proposition combining real-time GPS tracking with AI crowd management, clear target customer segments in public/private transit, and high technical team alignment for fast prototype development.

Areas of Improvement

Key areas for improvement include expanding initial sales and marketing efforts to acquire early pilot customers faster, optimizing hardware setup costs per bus, and refining long-term financial growth strategies.

Next Steps

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TIMELINE	GOALS	TEAM NEEDED	PHYSICAL RESOURCES NEEDED	FUNDS NEEDED
Months 1-3	Build functional working prototype and test real	2	Sensors	Rs. 2 Lakh
Months 4-6	Deploy pilot test on 10 buses and launch the pass	3	10 IoT GPS	3 lakh
Months 7-9	Validate pilot feedback, optimize crowd-counting	3	Co-worker	5 lakh
Months 10-12	Scale operations to 30+ buses	3	Yes	5 lakh

Venture Team



Name: J SHOPHANA

University/College: Sri Sairam engineering college

Major: Engineering

Key Skills: Communication

Role in the Venture: Leader

Keen on continuing with the venture:

Yes



Name: S.G Pushpasree

University/College: Sri Sairam engineering college

Major: Engineering

Key Skills: Communication

Role in the Venture: Member

Keen on continuing with the venture:

Yes



Name: B.Monisha

University/College: Sri Sairam engineering college

Major: Engineering

Key Skills: Communication

Role in the Venture: Member

Keen on continuing with the venture:

Yes

Current Mentors:
Faculty Advisors from the Electronics & Computer Science Department, along

Mentors Needed in these Areas:
Product development optimization for edge-AI hardware, pilot client acquisi

Venture Summary

OVERVIEW

We at TransitFlow AI are committed to transforming urban commuting by providing an AI-powered bus tracking and crowd management platform. Currently, we are developing an integrated solution that combines real-time GPS tracking with computer-vision crowd density analytics. Although we are in the early deployment phase, we are dedicated to reducing commuter wait times, preventing dangerous bus overcrowding, and helping transit operators optimize public transport efficiency.

Mission

Our mission is to make urban public transit safe, predictable, and efficient by leveraging AI-driven real-time tracking and intelligent crowd management to eliminate commuter waiting anxiety and prevent bus overcrowding

Social/Economic Relevance

Public transportation is the lifeline of daily commuting, but severe overcrowding and unpredictable schedules create daily stress, unsafe travel conditions, and lost productivity. Solving this reduces urban traffic congestion, enhances public safety, and builds accessible, sustainable smart cities for everyone.

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Thank You!

Our mission is to make urban public transit safe, predictable, and efficient by leveraging AI-driven real-time tracking and intelligent crowd management to eliminate commuter waiting anxiety and prevent bus overcrowding

